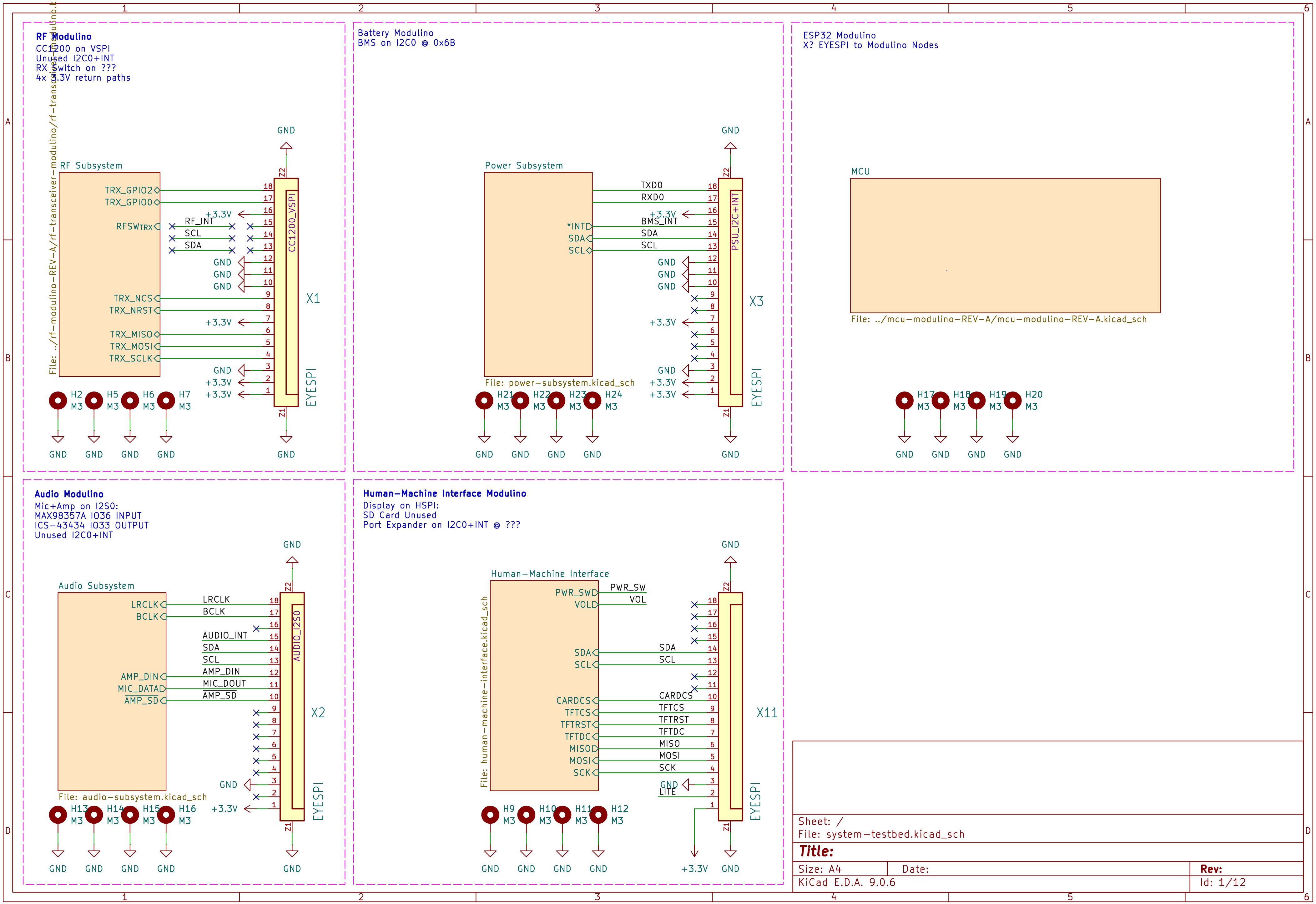
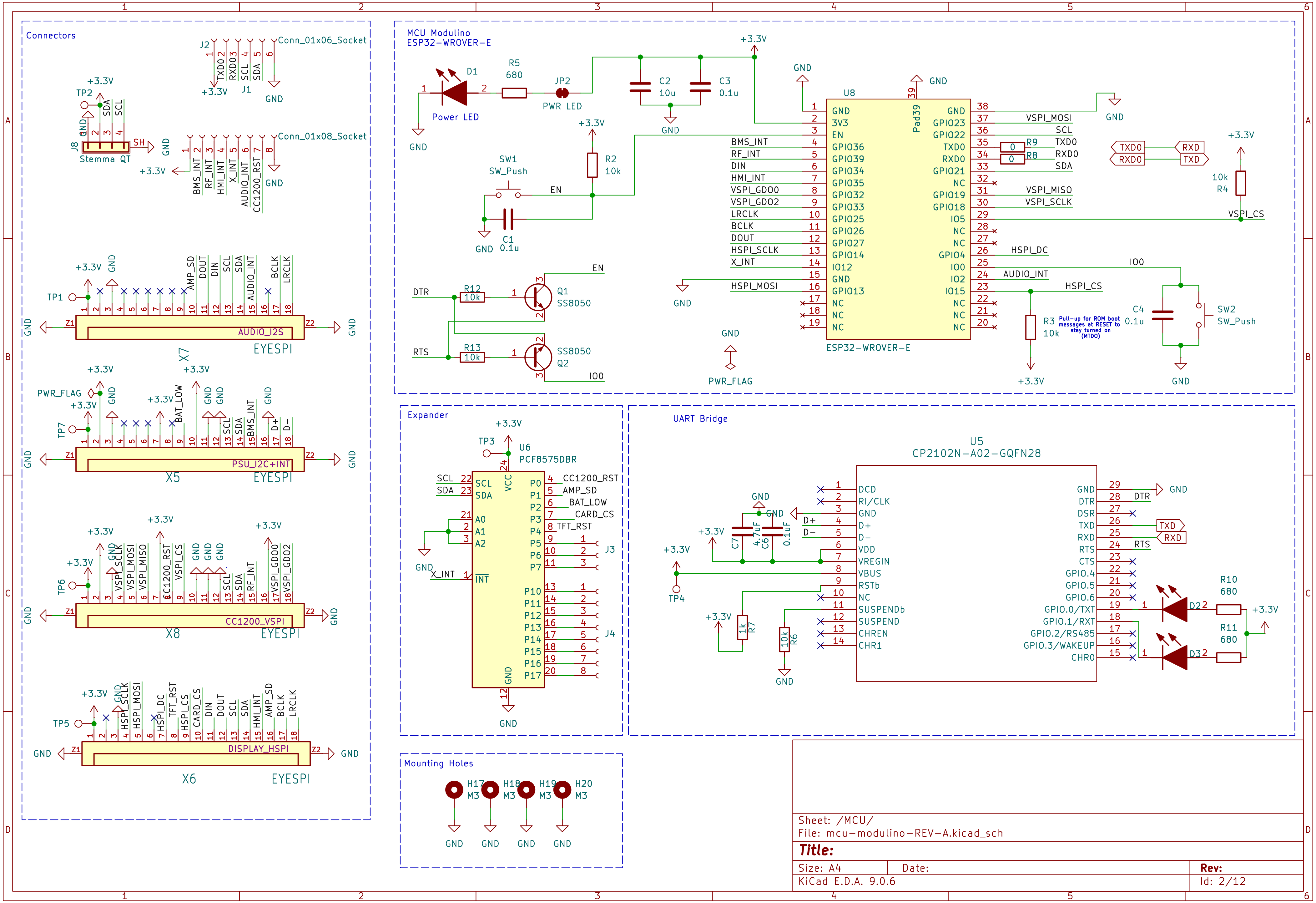
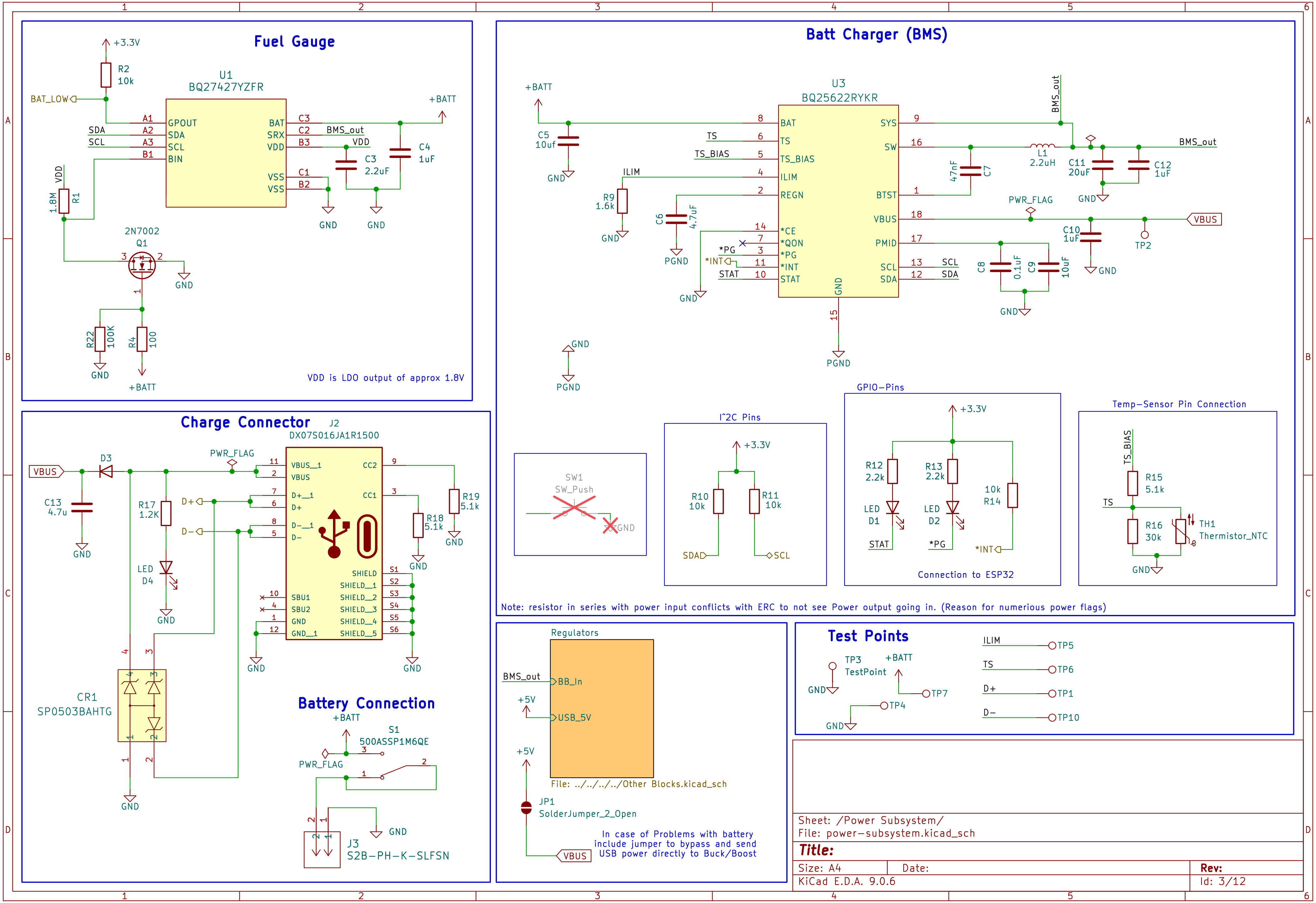

RF Modulino
CC1200 on VSPI
Unused I2C0+INT
RX Switch on ???
4x 3.3V return paths
RF Subsystem
File: ../rf-modulino-REV-A/rf-transceiver-modulino.kicad_sch
TRX_GPIO2
TRX_GPIO0
RFSWTRX
RF_INT
SCL
SDA
TRX_NCS
TRX_NRST
TRX_MISO
TRX_MOSI
TRX_SCLK
+3.3V
GND
CC1200_VSPI
X1
EYESPI
H2 M3
H5 M3
H6 M3
H7 M3
Battery Modulino
BMS on I2C0 @ 0x6B
Power Subsystem
File: power-subsystem.kicad_sch
*INT
SDA
SCL
TXD0
RXD0
BMS_INT
PSU_I2C+INT
X3
EYESPI
H21 M3
H22 M3
H23 M3
H24 M3
ESP32 Modulino
X? EYESPI to Modulino Nodes
MCU
File: ../mcu-modulino-REV-A/mcu-modulino-REV-A.kicad_sch
H17 M3
H18 M3
H19 M3
H20 M3
Audio Modulino
Mic+Amp on I2S0:
MAX98357A IO36 INPUT
ICS-43434 IO33 OUTPUT
Unused I2C0+INT
Audio Subsystem
File: audio-subsystem.kicad_sch
LRCLK
BCLK
AUDIO_INT
SDA
SCL
AMP_DIN
MIC_DATA
MIC_DOUT
AMP_SD
AUDIO_I2S0
X2
EYESPI
H13 M3
H14 M3
H15 M3
H16 M3
Human-Machine Interface Modulino
Display on HSPI:
SD Card Unused
Port Expander on I2C0+INT @ ???
Human-Machine Interface
File: human-machine-interface.kicad_sch
PWR_SW
VOL
SDA
SCL
CARDCS
TFTCS
TFTRST
TFTDC
MISO
MOSI
SCK
LITE
X11
EYESPI
H9 M3
H10 M3
H11 M3
H12 M3
Sheet: /
File: system-testbed.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 1/12


Connectors
J2
Conn_01x06_Socket
TXD0
RXD0
SCL
SDA
J1
Conn_01x08_Socket
BMS_INT
RF_INT
HMI_INT
X_INT
AUDIO_INT
CC1200_RST
TP2
J8
Stemma QT
SH
GND
+3.3V
TP1
X7
AUDIO_I2S
EYESPI
AMP_SD
DOUT
DIN
BCLK
LRCLK
PWR_FLAG
TP7
X5
PSU_I2C+INT
BAT_LOW
D+
D-
TP6
X8
CC1200_VSPI
VSPI_SCLK
VSPI_MOSI
VSPI_MISO
VSPI_CS
VSPI_GDO0
VSPI_GDO2
TP5
X6
DISPLAY_HSPI
HSPI_SCLK
HSPI_MOSI
HSPI_DC
TFT_RST
HSPI_CS
CARD_CS
MCU Modulino
ESP32-WROVER-E
D1
Power LED
R5
680
JP2
PWR LED
C2
10u
C3
0.1u
R2
10k
SW1
SW_Push
EN
C1
0.1u
U8
Pad39
GND
3V3
GPIO36
GPIO39
GPIO34
GPIO35
GPIO32
GPIO33
GPIO25
GPIO26
GPIO27
GPIO14
IO12
GPIO13
NC
GPIO23
GPIO22
TXD0
RXD0
GPIO21
GPIO19
GPIO18
IO5
GPIO4
IO0
IO2
IO15
HMI_INT
HSPI_SCLK
HSPI_MOSI
HSPI_DC
HSPI_CS
R9
0
R8
0
RXD
TXD
10k
R4
VSPI_CS
DTR
RTS
R12
10k
R13
10k
Q1
SS8050
Q2
SS8050
IO0
PWR_FLAG
R3
10k
Pull-up for ROM boot messages at RESET to stay turned on (MTDO)
C4
0.1u
SW2
SW_Push
Expander
U6
PCF8575DBR
TP3
VCC
A0
A1
A2
INT
P0
P1
P2
P3
P4
P5
P6
P7
P10
P11
P12
P13
P14
P15
P16
P17
J3
J4
UART Bridge
U5
CP2102N-A02-GQFN28
DCD
RI/CLK
VDD
VREGIN
VBUS
RSTb
SUSPENDb
SUSPEND
CHREN
CHR1
DSR
CTS
GPIO.4
GPIO.5
GPIO.6
GPIO.0/TXT
GPIO.1/RXT
GPIO.2/RS485
GPIO.3/WAKEUP
CHR0
C7
4.7uF
C6
0.1uF
R7
1k
R6
10k
TP4
D2
D3
R10
680
R11
680
Mounting Holes
H17
H18
H19
H20
M3
Sheet: /MCU/
File: mcu-modulino-REV-A.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 2/12


Fuel Gauge
+3.3V
R2
10k
BAT_LOW
U1
BQ27427YZFR
GPOUT
SDA
SCL
BIN
BAT
SRX
VDD
VSS
VSS
A1
A2
A3
B1
C3
C2
B3
C1
B2
BMS_out
VDD
+BATT
C3
2.2uF
C4
1uF
GND
GND
VDD
R1
1.8M
2N7002
Q1
GND
R22
100K
R4
100
GND
+BATT
VDD is LDO output of approx 1.8V
Batt Charger (BMS)
U3
BQ25622RYKR
+BATT
C5
10uf
GND
BAT
TS
TS_BIAS
ILIM
REGN
*CE
*QON
*PG
*INT
STAT
SYS
SW
BTST
VBUS
PMID
SCL
SDA
GND
TS
TS_BIAS
ILIM
R9
1.6k
GND
C6
4.7uF
PGND
*PG
*INT
STAT
GND
BMS_out
L1
2.2uH
C7
47nF
C11
20uF
C12
1uF
BMS_out
GND
PWR_FLAG
VBUS
C10
1uF
TP2
GND
C8
0.1uF
C9
10uF
GND
SCL
SDA
PGND
GND
PGND
SW1
SW_Push
GND
I^2C Pins
+3.3V
R10
10k
R11
10k
SDA
SCL
GPIO-Pins
+3.3V
R12
2.2k
R13
2.2k
R14
10k
LED
D1
LED
D2
STAT
*PG
*INT
Connection to ESP32
Temp-Sensor Pin Connection
TS_BIAS
R15
5.1k
TS
R16
30k
TH1
Thermistor_NTC
GND
Note: resistor in series with power input conflicts with ERC to not see Power output going in. (Reason for numerious power flags)
Charge Connector
J2
DX07S016JA1R1500
VBUS
D3
C13
4.7u
GND
PWR_FLAG
R17
1.2K
LED
D4
GND
D+
D-
VBUS_1
VBUS
D+_1
D+
D-_1
D-
SBU1
SBU2
GND
GND_1
CC2
CC1
SHIELD
SHIELD_1
SHIELD_2
SHIELD_3
SHIELD_4
SHIELD_5
R19
5.1k
R18
5.1k
GND
GND
GND
GND
CR1
SP0503BAHTG
GND
Battery Connection
+BATT
S1
500ASSP1M6QE
PWR_FLAG
J3
S2B-PH-K-SLFSN
GND
Regulators
BMS_out
BB_In
+5V
USB_5V
File: ../../../../Other Blocks.kicad_sch
+5V
JP1
SolderJumper_2_Open
VBUS
In case of Problems with battery
include jumper to bypass and send
USB power directly to Buck/Boost
Test Points
TP3
TestPoint
GND
+BATT
TP7
GND
TP4
ILIM
TP5
TS
TP6
D+
TP1
D-
TP10
Sheet: /Power Subsystem/
File: power-subsystem.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
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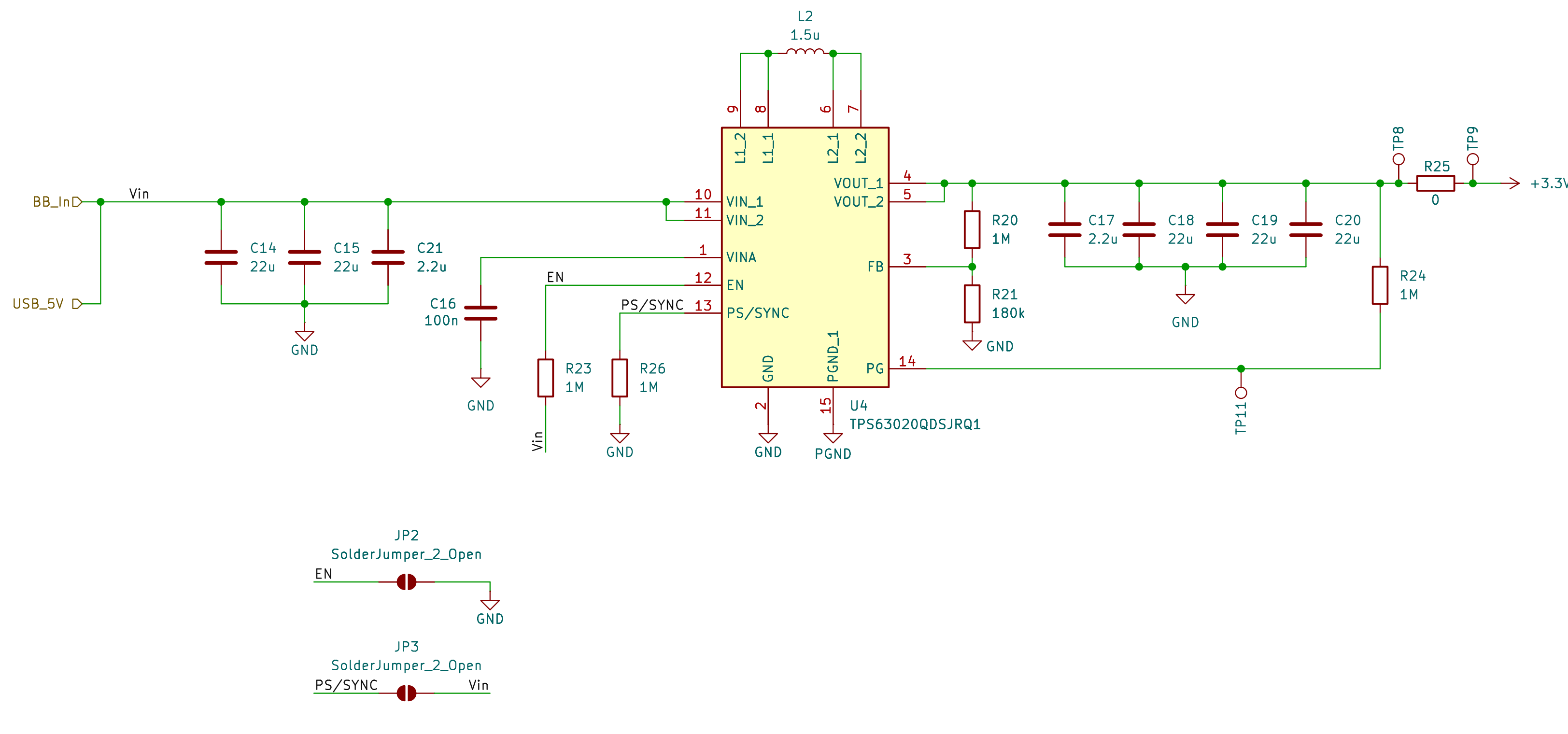
C: DC–DC Conversion (Voltage Regulation)

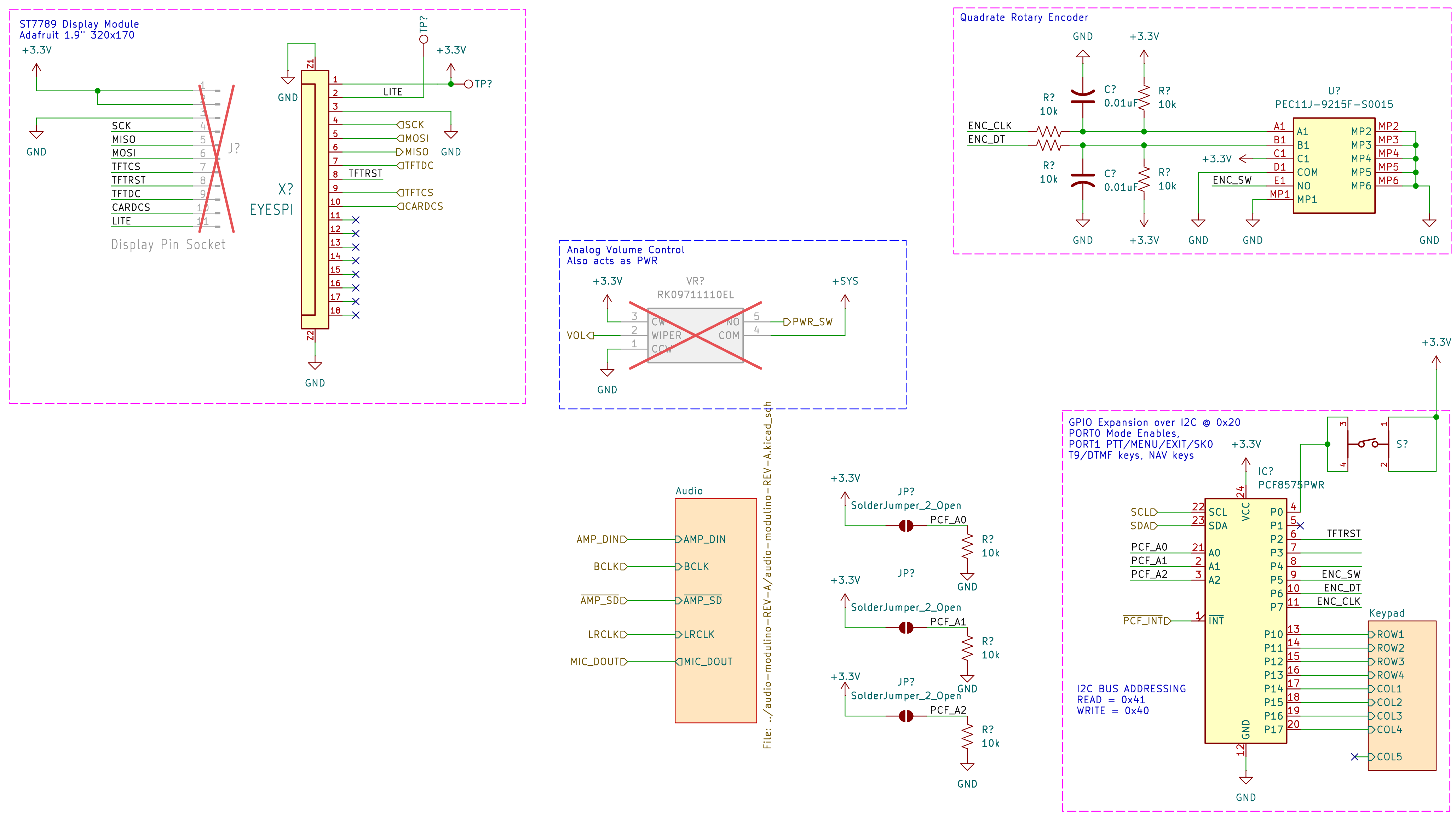


Sheet: /Power Subsystem/Regulators/
File: Other Blocks.kicad_sch

Title:

Size: B | Date: | **Rev:**

KiCad E.D.A. 9.0.6 | Id: 4/12

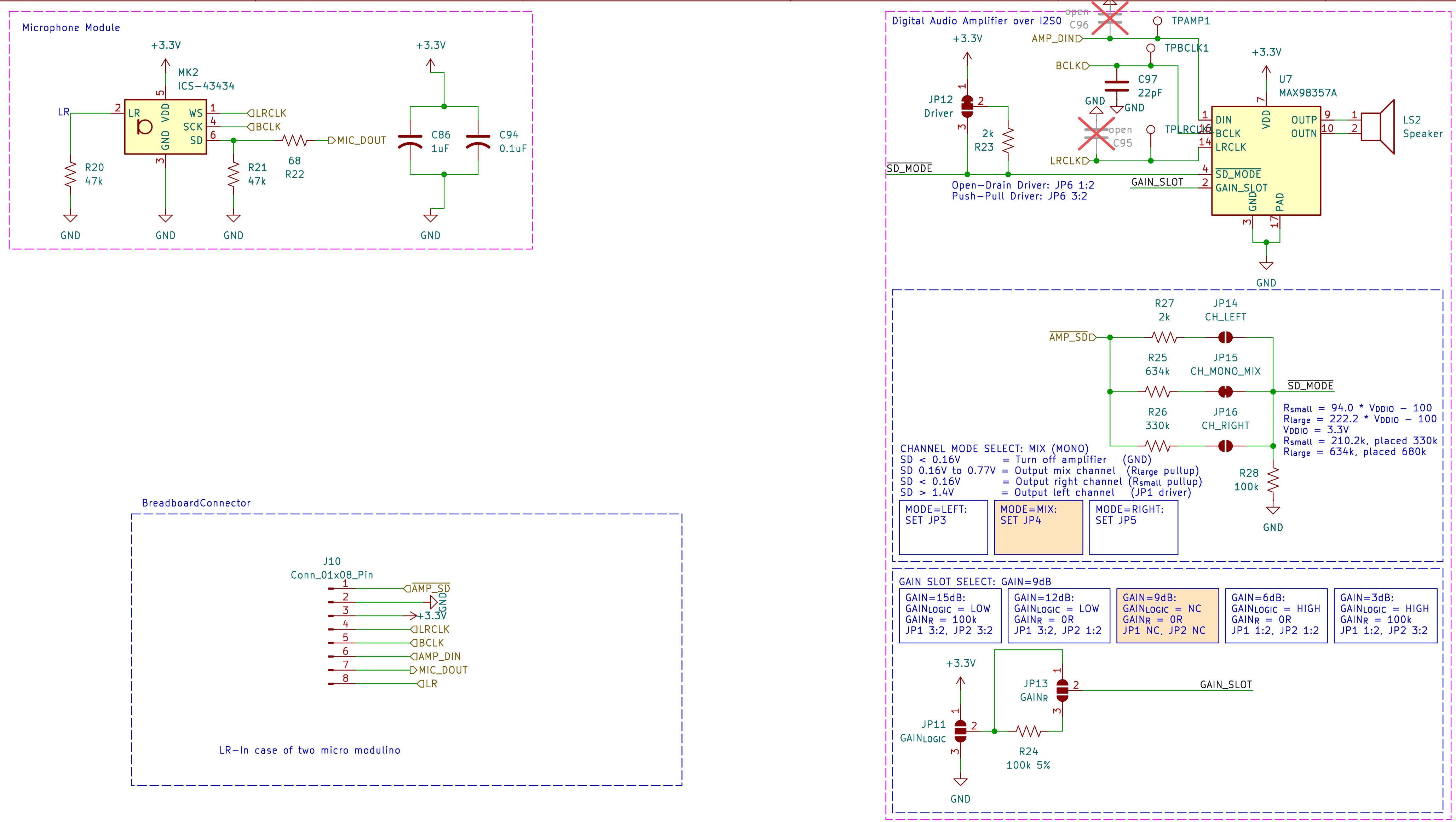


Sheet: /Human-Machine Interface/
File: human-machine-interface.kicad_sch

Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.6 | Id: 5/12



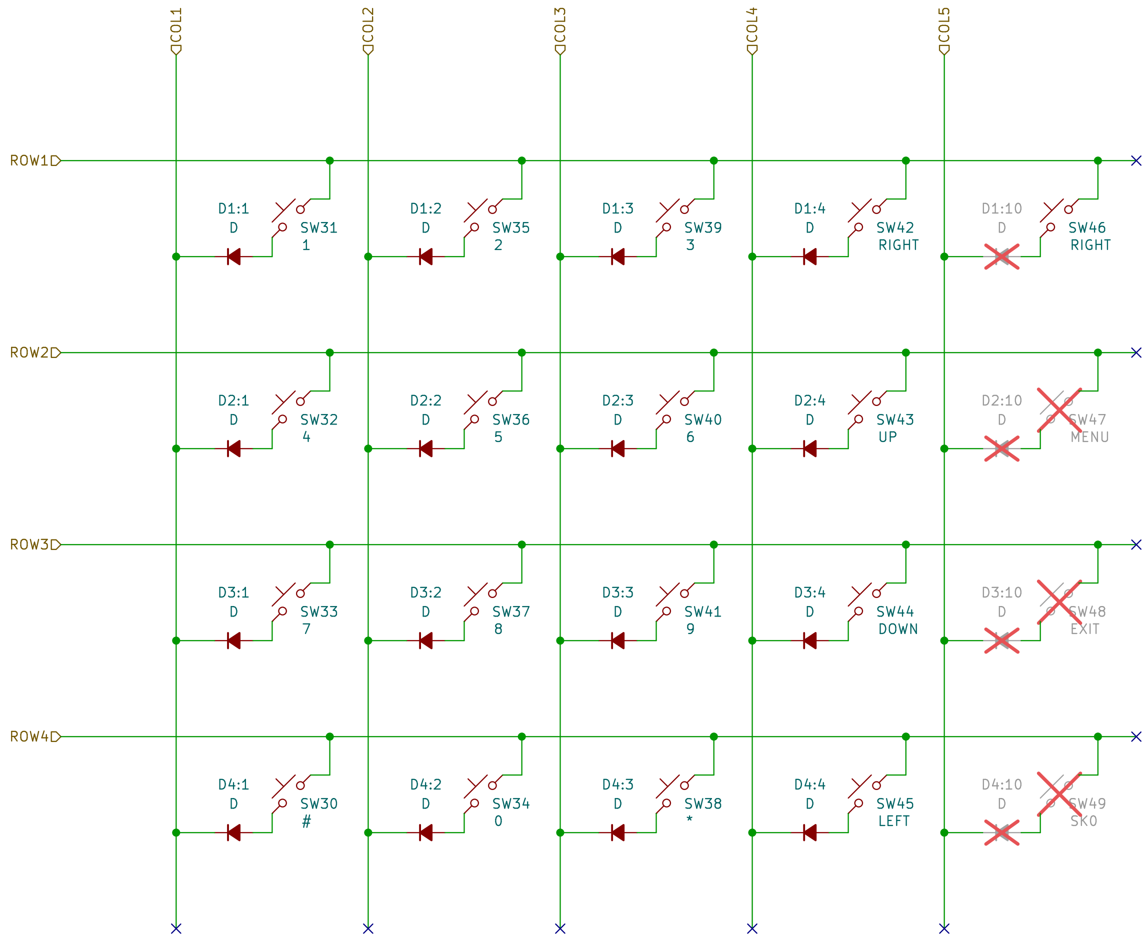
Sheet: /Human-Machine Interface/Audio/
File: audio-modulino-REV-A.kicad_sch

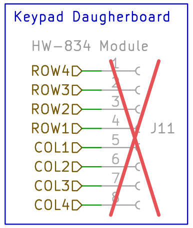
Title:

Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.6 | Id: 6/12

Keypad Matrix
Application Note: The ROW pins get pulled up by the GPIO expander, with one ROW being pulled down.
The scanner relies on the COL pins being read in as a logic swing to low.
Remarked as Domino Logic, a rail-to-rail logic swing occurs.

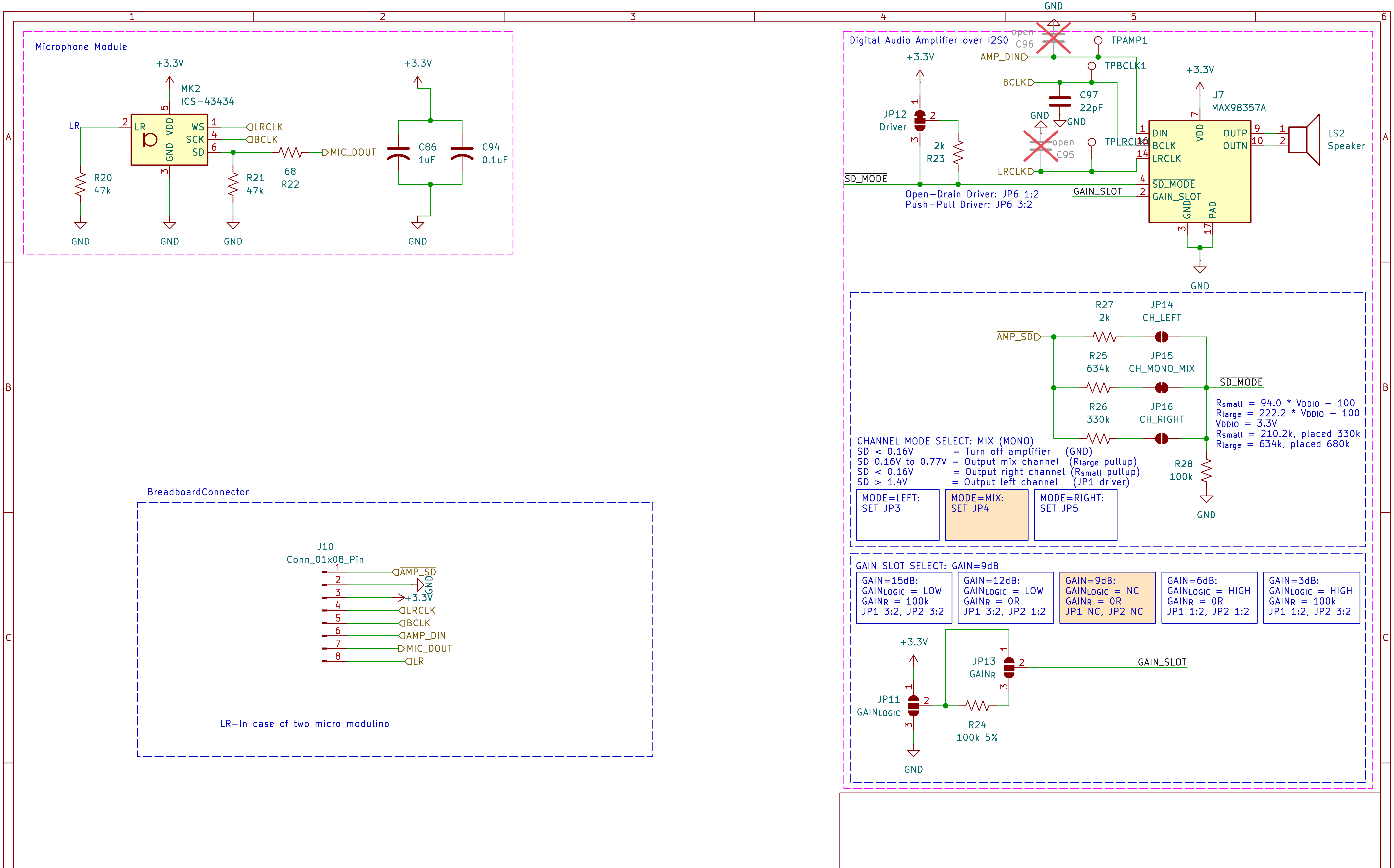




Sheet: /Human-Machine Interface/Keypad/
File: keypad.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 7/12

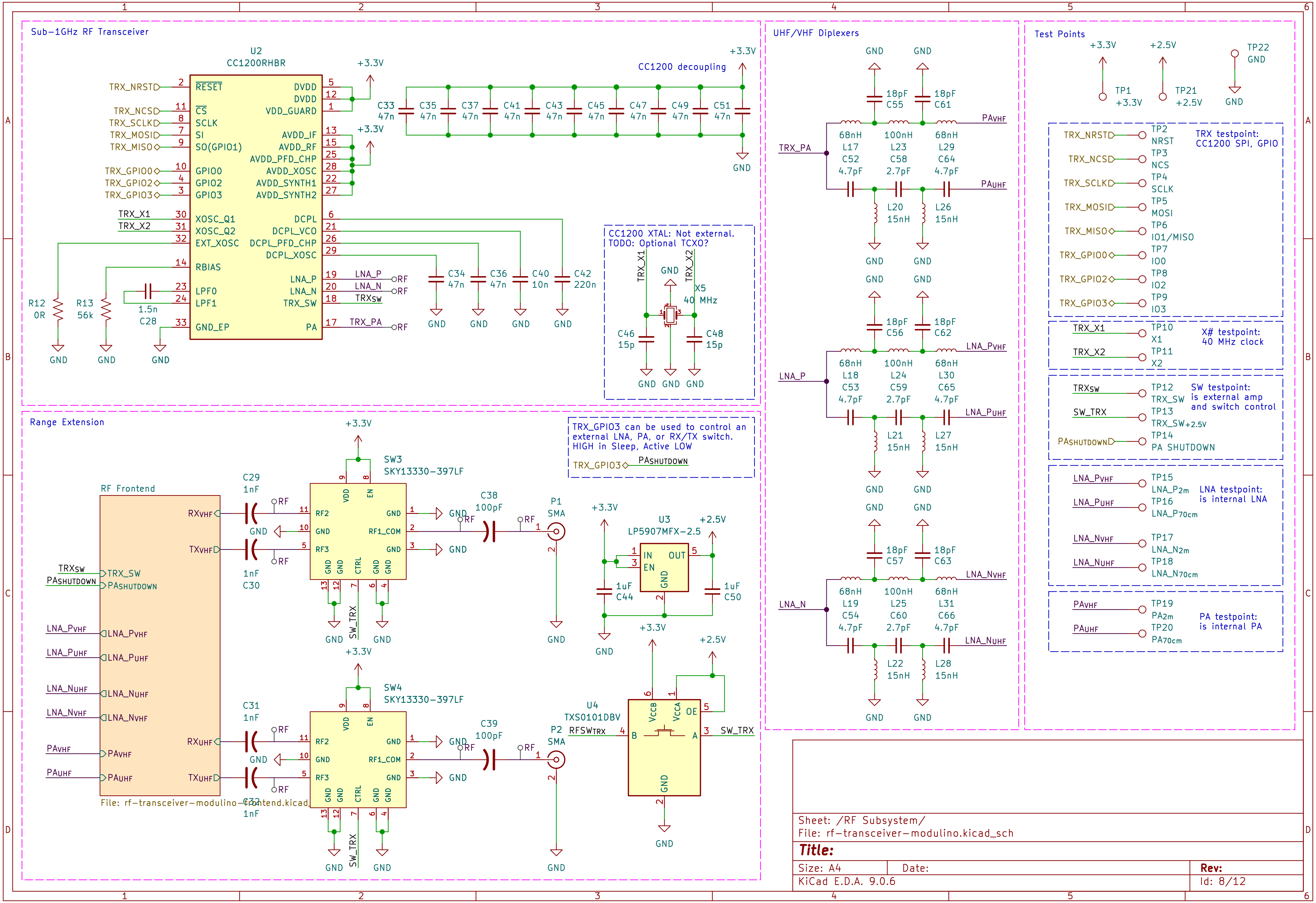


Sheet: /Audio Subsystem/
File: audio-subsystem.kicad_sch

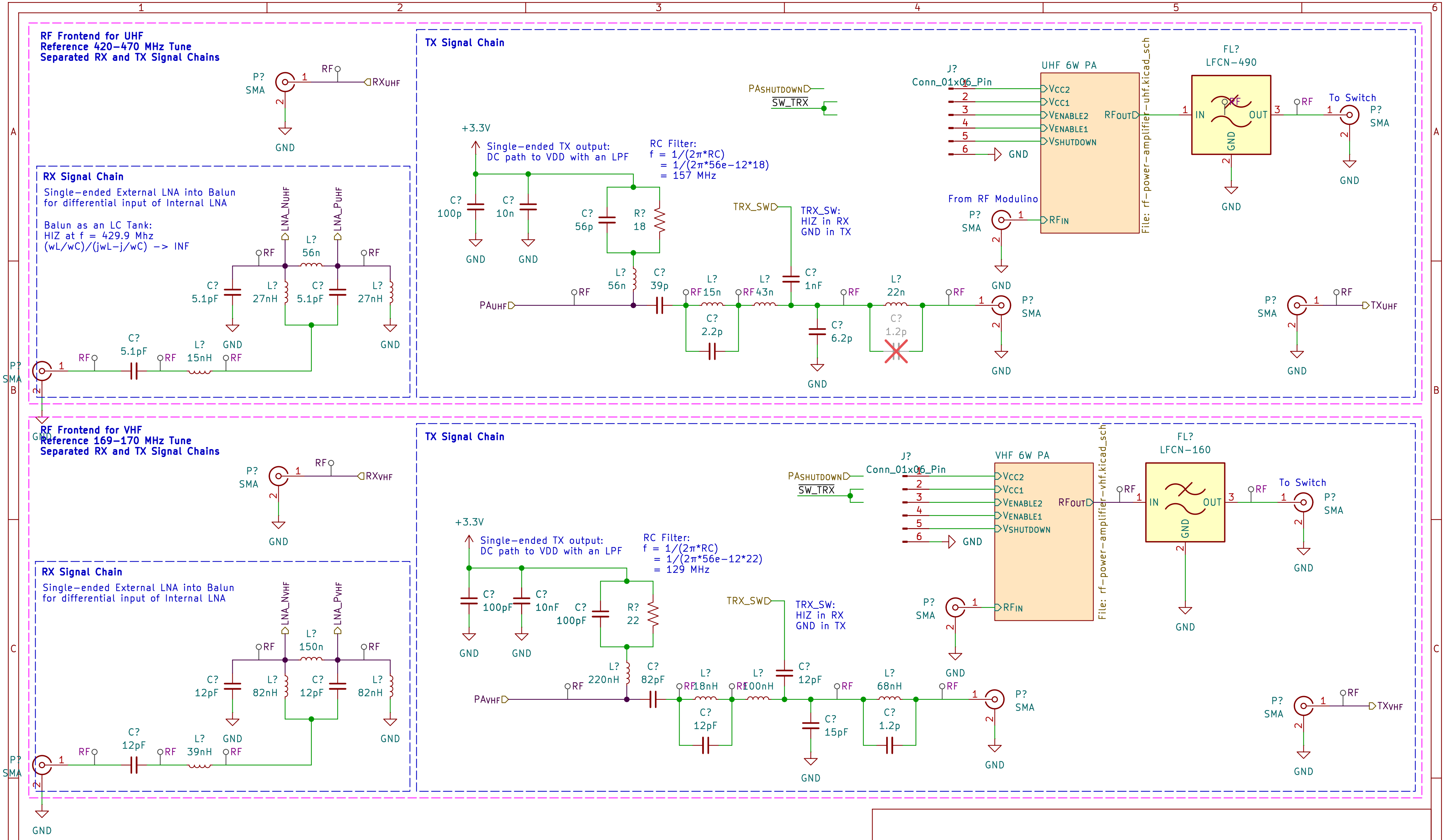
Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 9.0.6 Id: 6/12



Sub-1GHz RF Transceiver
U2
CC1200RHBR
RESET
CS
SCLK
SI
SO(GPIO1)
GPIO0
GPIO2
GPIO3
XOSC_Q1
XOSC_Q2
EXT_XOSC
RBIAS
LPF0
LPF1
GND_EP
DVDD
DVDD
VDD_GUARD
AVDD_IF
AVDD_RF
AVDD_PFD_CHP
AVDD_XOSC
AVDD_SYNTH1
AVDD_SYNTH2
DCPL
DCPL_VCO
DCPL_PFD_CHP
DCPL_XOSC
LNA_P
LNA_N
TRX_SW
PA
TRX_NRST
TRX_NCS
TRX_SCLK
TRX_MOSI
TRX_MISO
TRX_GPIO0
TRX_GPIO2
TRX_GPIO3
TRX_X1
TRX_X2
TRX_PA
R12 0R
R13 56k
1.5n C28
+3.3V
GND
CC1200 decoupling
C33 47n
C35 47n
C37 47n
C41 47n
C43 47n
C45 47n
C47 47n
C49 47n
C51 47n
C34 47n
C36 47n
C40 10n
C42 220n
CC1200 XTAL: Not external.
TODO: Optional TCXO?
X5
40 MHz
C46 15p
C48 15p
UHF/VHF Diplexers
18pF C55
18pF C61
PAVHF
PAUHF
68nH L17
100nH L23
68nH L29
C52 4.7pF
C58 2.7pF
C64 4.7pF
L20 15nH
L26 15nH
18pF C56
18pF C62
LNA_PVHF
LNA_PUHF
68nH L18
100nH L24
68nH L30
C53 4.7pF
C59 2.7pF
C65 4.7pF
L21 15nH
L27 15nH
18pF C57
18pF C63
LNA_NVHF
LNA_NUHF
68nH L19
100nH L25
68nH L31
C54 4.7pF
C60 2.7pF
C66 4.7pF
L22 15nH
L28 15nH
Test Points
+3.3V TP1
+2.5V TP21
TP22 GND
TRX testpoint:
CC1200 SPI, GPIO
TP2 NRST
TP3 NCS
TP4 SCLK
TP5 MOSI
TP6 IO1/MISO
TP7 IO0
TP8 IO2
TP9 IO3
X# testpoint:
40 MHz clock
TP10 X1
TP11 X2
SW testpoint:
is external amp
and switch control
TP12 TRX_SW
TP13 TRX_SW+2.5V
TP14 PA SHUTDOWN
TRXSW
SW_TRX
PASHUTDOWN
LNA testpoint:
is internal LNA
TP15 LNA_P2m
TP16 LNA_P70cm
TP17 LNA_N2m
TP18 LNA_N70cm
PA testpoint:
is internal PA
TP19 PA2m
TP20 PA70cm
Range Extension
TRX_GPIO3 can be used to control an
external LNA, PA, or RX/TX switch.
HIGH in Sleep, Active LOW
RF Frontend
RXVHF
TXVHF
RXUHF
TXUHF
File: rf-transceiver-modulino-frontend.kicad_sch
C29 1nF
C30 1nF
C31 1nF
C32 1nF
SW3
SKY13330-397LF
SW4
SKY13330-397LF
RF2
RF3
RF1_COM
VDD
EN
CTRL
C38 100pF
C39 100pF
P1 SMA
P2 SMA
U3
LP5907MFX-2.5
IN
EN
OUT
1uF C44
1uF C50
+2.5V
U4
TXS0101DBV
RFSWTRX
VCCB
VCCA
OE
Sheet: /RF Subsystem/
File: rf-transceiver-modulino.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 8/12



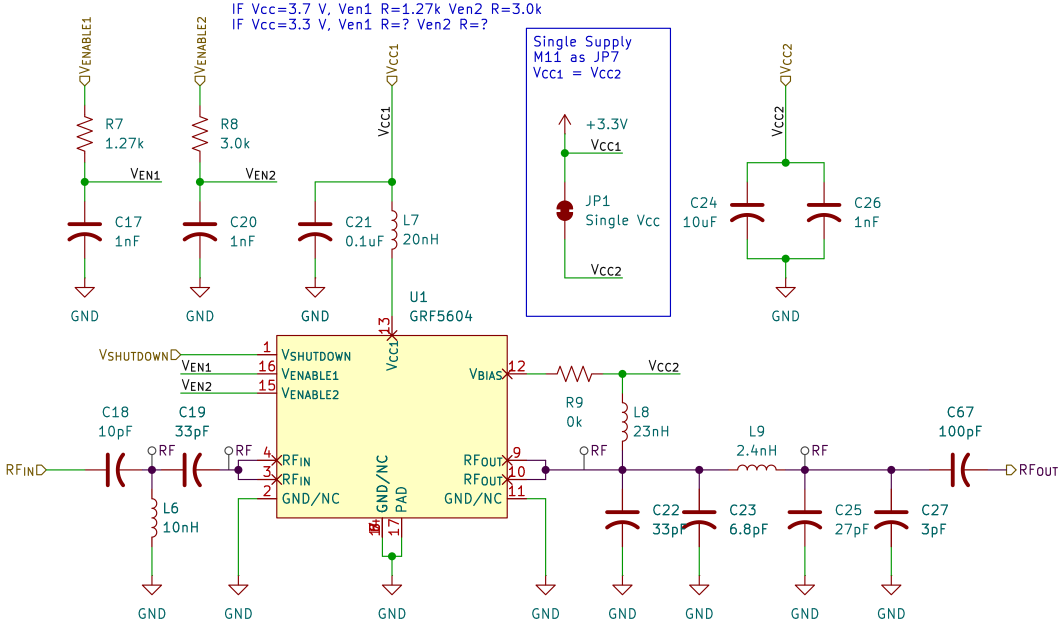
Sheet: /RF Subsystem/RF Frontend/
File: rf–transceiver–modulino–frontend.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 9/12

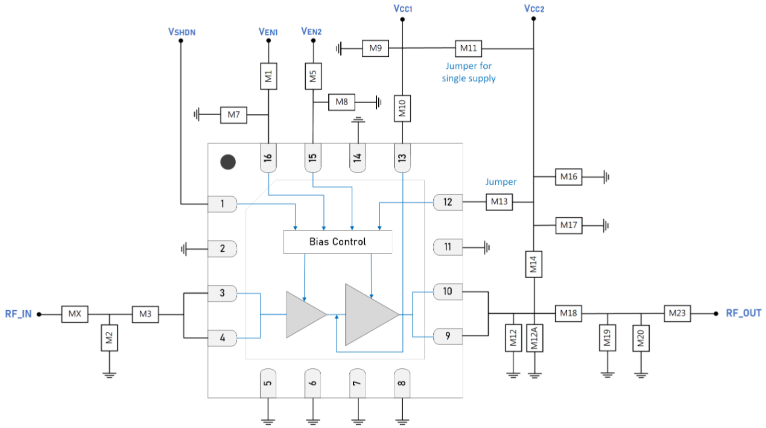
RF Amplifier for UHF
Reference 397–470 MHz Tune

IF Vcc=5.0 V, Ven1 R=1.96k Ven2 R=6.04k
IF Vcc=3.7 V, Ven1 R=1.27k Ven2 R=3.0k
IF Vcc=3.3 V, Ven1 R=? Ven2 R=?



Ref	Eval. Ref	450–470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 µF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

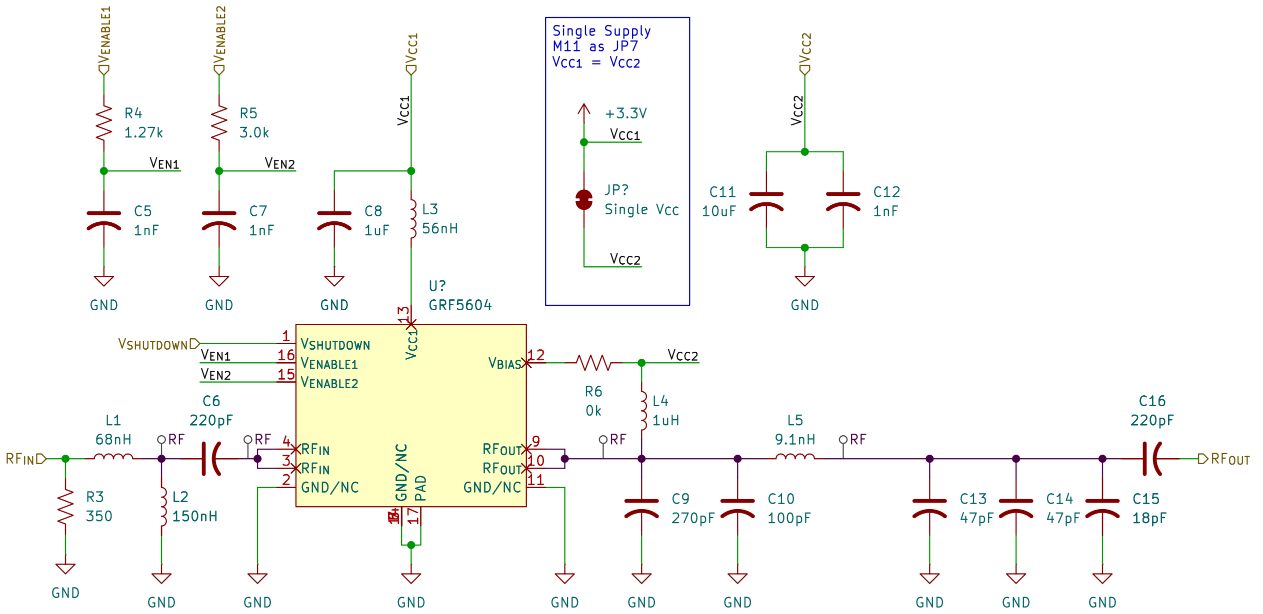
Sheet: /RF Subsystem/RF Frontend/VHF 6W PA/
File: rf–power–amplifier–vhf.kicad_sch

Title:

Size: A4 Date: **Rev:**
KiCad E.D.A. 9.0.6 Id: 6/12

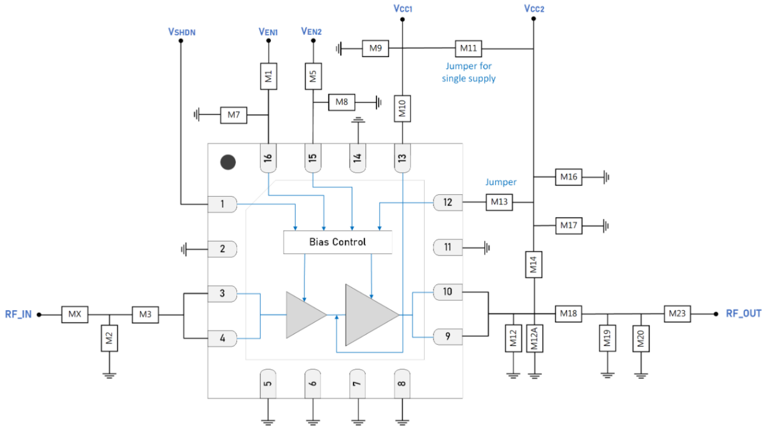
RF Amplifier for VHF
Reference 135–175 MHz Tune

IF Vcc=5.0 V, Ven1 R=1.96k Ven2 R=6.04k
IF Vcc=3.7 V, Ven1 R=1.27k Ven2 R=3.0k
IF Vcc=3.3 V, Ven1 R=? Ven2 R=?



Ref	Eval. Ref	450–470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 µF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/UHF 6W PA/
File: rf–power–amplifier–uhf.kicad_sch

Title:

Size: A4 Date:
KiCad E.D.A. 9.0.6

Rev:
Id: 10/12